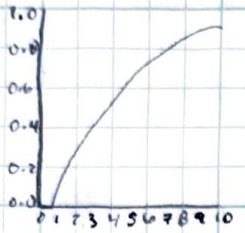
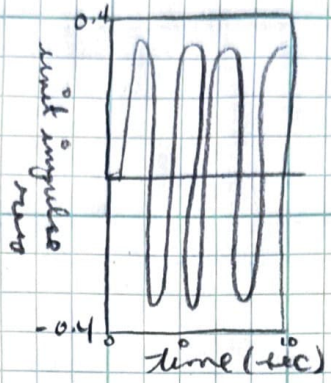
Seth Ricks ECEN 4710 10/19/2020 PSOS: 4.46(a)(b), 4.48(a)(b) (show that it works), 4.49(b)

(4.46) Unit-step response of paper-making shown, where input to system is stock flow onto wire and output is basis weight (thickness). Time delay & slope of transient response may be determined from figure.

1/4

Step response





a.) Find P, PI, & PID parameters using Ziegler-Nichols transient response method:

$D_c(s) = K_p + \frac{K_I}{s} + K_D s$ $\quad R \approx \frac{0.2}{0.5} = 0.4$

$L \approx 0.5$

P controller: $k_p = \frac{1}{RL} = 5$ $\boxed{K_p = 5}$

PI controller: $k_p = \frac{0.9}{RL} = 4.5$ $\boxed{K_p = 4.5 ; K_I = 2.7}$

$D_c(s) = k_p + \frac{k_p}{T_I} s$

$K_I = \frac{K_p}{T_I} = \frac{K_p}{L/0.3} = \frac{K_p \cdot 0.3}{L} = 2.7$

PID controller:

$D_c(s) = k_p + \frac{k_p}{T_I} s + k_p T_D s$ $\quad K_p = \frac{1.2}{RL} = 6$ $\boxed{K_p = 6 ; K_I = 6 ; K_D = 1.5}$

$K_I = \frac{K_p}{T_I} = \frac{K_p}{2L} = 6$

$K_D = K_p T_D = K_p(0.5L) = 1.5$

b.) Using proportional feedback control. When gain $K_u = 8.556$, system marginally stable. Determine P, PI, PID controller parameters w/ ultimate sensitivity method.

2/4

$K_u = 8.556$
$P_u \approx 2.25$ sec

P controller: $K_p = 0.5K_u = 4.278$ $\boxed{K_p = 4.278}$

PI controller: $K_p = 0.45K_u = 3.8502$ $\boxed{K_p = 3.850 \; ; \; K_I = 2.0537}$

$D_c(s) = K_p + \frac{K_p}{T_I}s$ $K_I = \frac{K_p}{T_I} = \frac{K_p}{\frac{P_u}{1.2}} = \frac{1.2K_p}{P_u} = 2.053$

PID controller:

$D_c(s) = K_p + \frac{K_p}{T_I}s + K_pT_Ds$ $K_p = 0.6K_u = 5.134$ $\boxed{K_p = 5.134 \; ; \; K_I = 4.563 \; ; \; K_D = 1.444}$

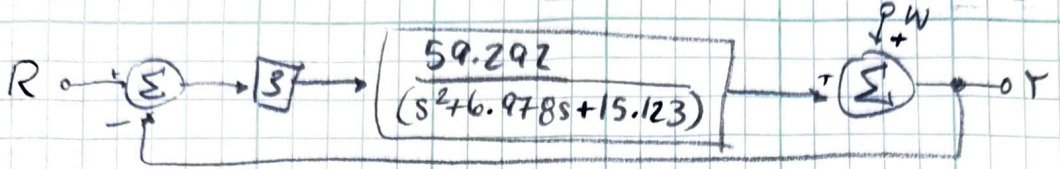
$K_I = \frac{K_p}{T_I} = \frac{K_p}{0.5P_u} = 4.563$

$K_D = K_pT_D = K_p(0.125P_u) = 1.444$

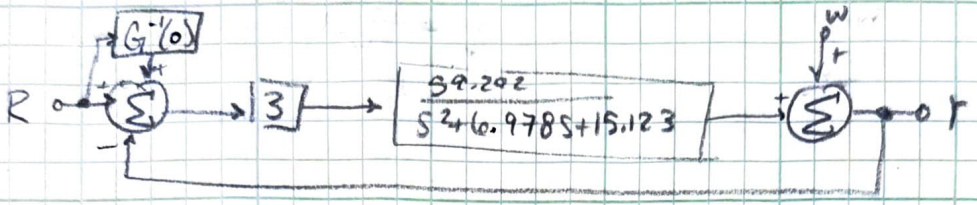
7.48 Consider the DC motor speed-control system shown, with proportional control.

a.) Add feedforward control to eliminate steady-state error for a step reference input.

b.) Also add feedforward control to eliminate the effect of a constant output disturbance signal, w, on the output of the system.

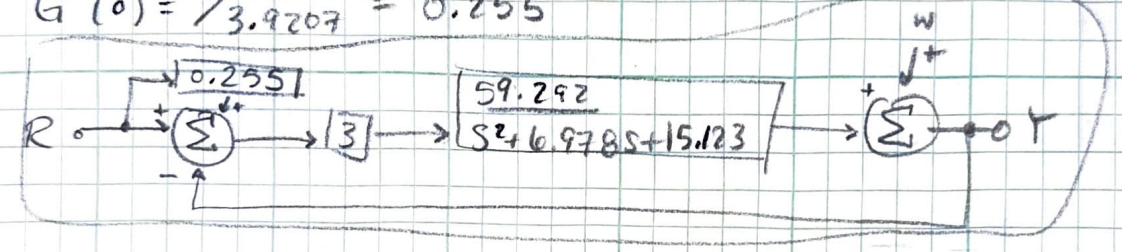


a.)



$$G(0) = \frac{59.292}{15.123} = 3.9207$$

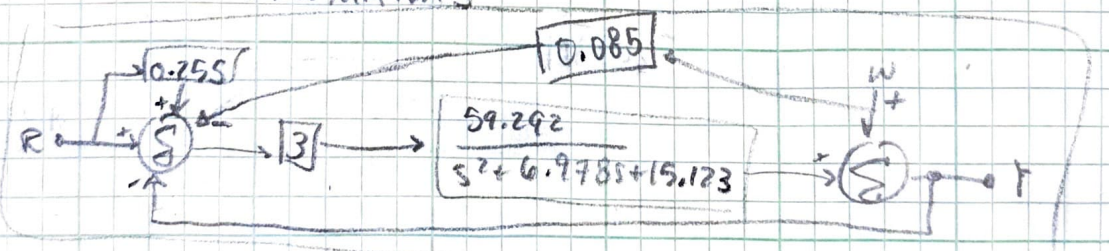
$$G^{-1}(0) = 1/3.9207 = 0.255$$



Original: $H(s) = \frac{K_p G}{1 + K_p G}$; new: $H(s) = (1 + K_{ff})\left(\frac{K_p G}{1 + K_p G}\right)$

* see MATLAB

b.)



$K_p G(R + K_{ff} R + -Y - K_{fw} W) + W = Y$

$K_p G R + K_p K_{ff} G R - K_p G Y + K_p K_{fw} W G + W = Y$

$K_p G R + K_p K_{ff} G R - K_p K_{fw} W G + W = Y + K_p G Y = Y(1 + K_p G)$

$R[K_p G + K_p K_{ff} G] + W(1 - K_p K_{fw} G) = Y(1 + K_p G)$

$1 - K_p K_{fw} G = 0$

$K_{fw} = \frac{1}{K_p G} \rightarrow \frac{G^{-1}(0)}{K_p}$

$$Y = \frac{G K_p (1 + K_{ff})}{1 + K_p G} R(s) + \frac{1 - K_p K_{fw} G}{1 + K_p G} W(s)$$

* see Matlab

Problem 4.48

Consider the DC motor speed-control system shown in Fig. 4.60 with proportional control.

(a) Add feedforward control to eliminate the steady-state tracking error for a step reference input.

(b) Also add feedforward control to eliminate the effect of a constant output disturbance signal, w, on the output of the system.

a.)

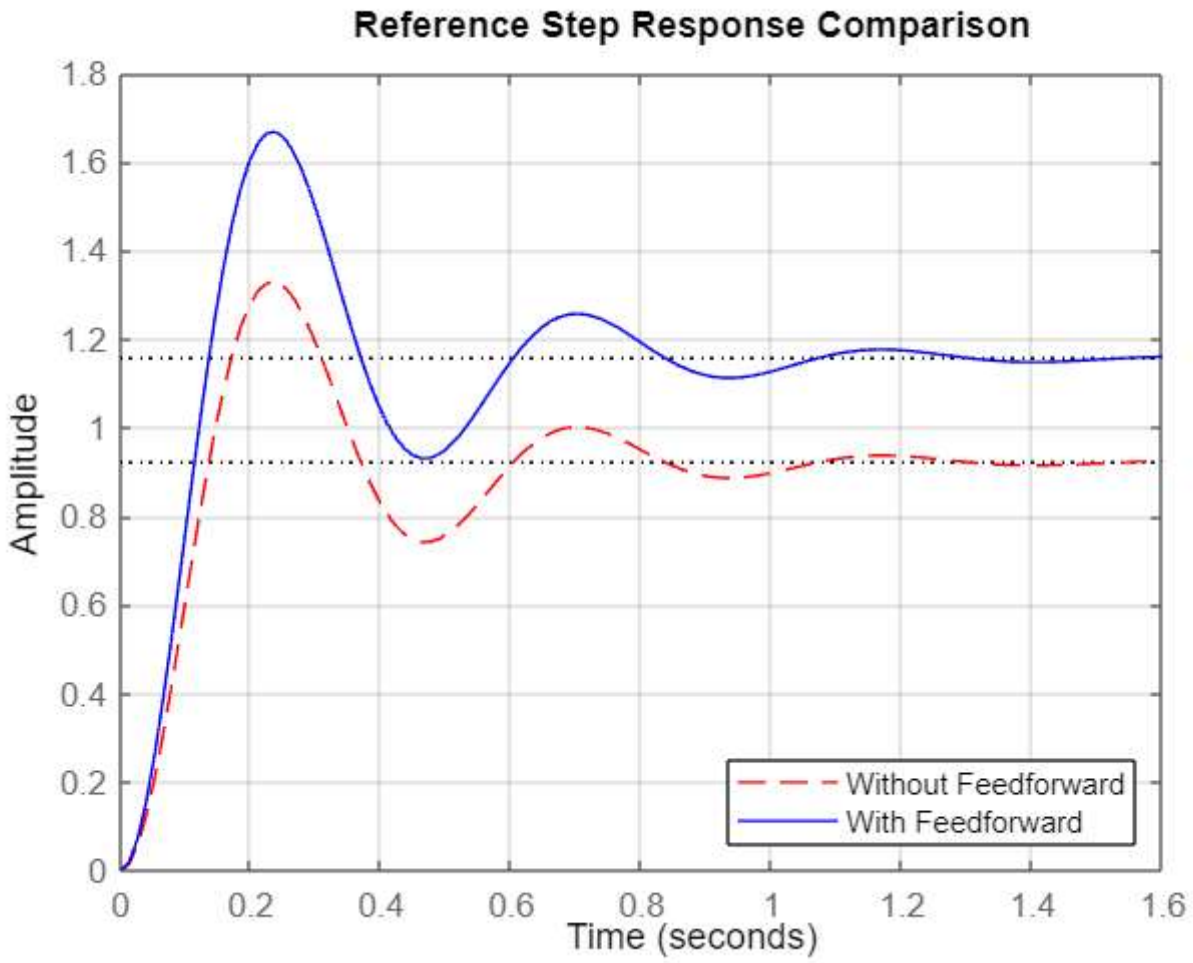
```
G = tf(59.292, [1, 6.978, 15.123]);

Kp = 3;

K_ff = 15.123 / 59.292;

sys_no_ff = (Kp * G) / (1 + Kp * G);
sys_ff = (1+K_ff) * sys_no_ff;

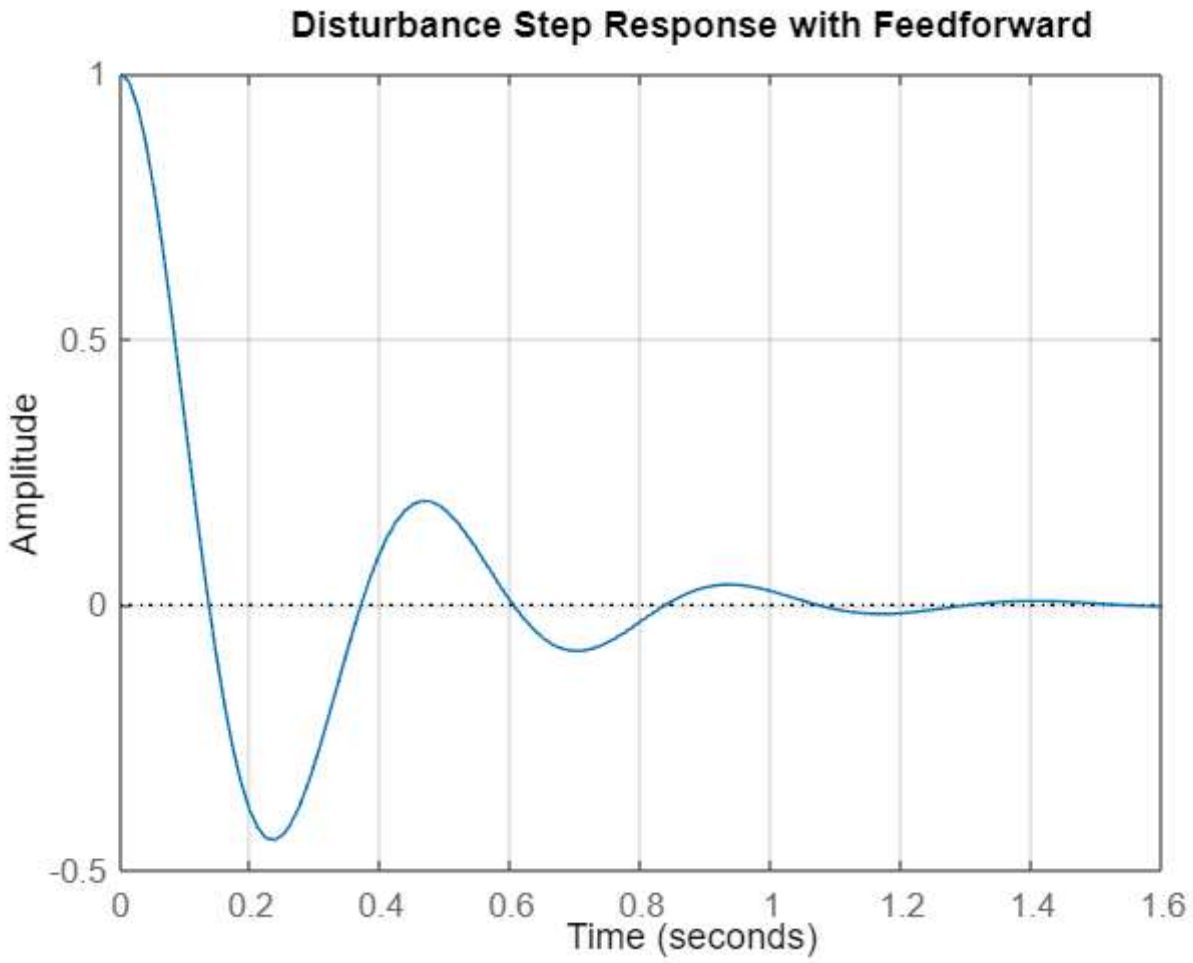
figure;
step(sys_no_ff, 'r--', sys_ff, 'b-');
legend('Without Feedforward', 'With Feedforward', 'Location', 'Southeast');
title('Reference Step Response Comparison');
grid on;
```



b.)

```
K_fw = K_ff / Kp;
sys_dist = (1 - Kp * K_fw * G) / (1 + Kp * G);

figure;
step(sys_dist);
title('Disturbance Step Response with Feedforward');
grid on;
```




```
disp('Steady-state gain of disturbance response:');
```

```
Steady-state gain of disturbance response:
     9.2040e-018
```

(4.49) Compute the discrete equivalents for the following possible controllers using the trapezoid rule. Let $T_s = 0.05$ sec.

b.) $D_c(s) = 2\,\frac{s+2}{s+4}$

$D_d(z) = D_c\left(\frac{2}{T_s}\,\frac{z-1}{z+1}\right)$ ✱ Notecard! ↳ cont. to discrete

$$D_d(z) = 2\left[\frac{\frac{2(z-1)}{T_s(z+1)} + 2}{\frac{2(z-1)}{T_s(z+1)} + 4}\right]$$

$$= \frac{\frac{4(z-1)}{T_s(z+1)} + 4}{\frac{2(z-1)}{T_s(z+1)} + 4} = \frac{\frac{4(z-1)}{T_s(z+1)} + \frac{4T_s(z+1)}{T_s(z+1)}}{\frac{2(z-1)}{T_s(z+1)} + \frac{4T_s(z+1)}{T_s(z+1)}}$$

$$= \frac{4(z-1) + 4T_s(z+1)}{2(z-1) + 4T_s(z+1)}$$

$$= \frac{4z - 4 + 4T_s z + 4T_s}{2z - 2 + 4T_s z + 4T_s}$$

$$= \frac{4z - 4 + 4(0.05)z + 4(0.05)}{2z - 2 + 4(0.05)z + 4(0.05)}$$

$$= \frac{4z - 4 + 0.2z + 0.2}{2z - 2 + 0.2z + 0.2} = \frac{4.2z - 3.8}{2.2z - 1.8}$$

$$D_d(z) = \frac{U(z)}{E(z)} = \frac{4.2z - 3.8}{2.2z - 1.8} * \frac{5}{5}$$

$$D_d(z) = \frac{21z - 19}{11z - 9}$$

$(11z - 9)U(z) = (21z - 19)E(z)$

$11zU(z) - 9U(z) = 21zE(z) - 19E(z)$

$11u(k+1) - 9u(k) = 21e(k+1) - 19e(k)$

$11u(k) - 9u(k-1) = 21e(k) - 19e(k-1)$

$$u(k) = \frac{9}{11}u(k-1) + \frac{21}{11}e(k) - \frac{19}{11}e(k-1)$$

✱ $z \cdot U(z) \Leftrightarrow u(k+1)$
✱ k = discrete time index
✱ z = transform domain